

TOBERMORY, ON, Canada

WMO: 717670

Lat: 45.2239N

Lon: 81.6350W

Elev: 214

StdP: 98.79

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.5	-16.9	-22.6	0.5	-18.7	-20.3	0.6	-16.2	10.6	-4.5	9.4	-4.6	3.2	0	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.3	25.5	19.6	24.3	19.2	23.1	18.7	21.1	23.8	20.4	22.9	19.7	22.0	3.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	15.2	22.6	19.5	14.6	22.0	18.8	14.0	21.3	62.4	23.8	59.8	23.0	57.3	22.1	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.8	7.0	DB	-22.9	28.1	4.2	1.6	-26.0	29.2	-28.4	30.1	-30.8	31.0	-33.8	32.2	(4)
(5)			WB	-23.2	22.5	4.2	1.1	-26.2	23.3	-28.6	24.0	-31.0	24.6	-34.0	25.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-6.2	-6.3	-2.4	3.6	10.2	14.8	18.5	18.4	15.0	9.1	2.8	-2.1	(6)
(7)		DBStd	9.84	5.89	5.53	5.50	4.18	3.89	2.99	2.55	2.50	3.61	4.05	4.38	4.50	(7)
(8)		HDD10.0	2252	502	457	385	196	46	2	0	0	4	67	219	374	(8)
(9)		HDD18.3	4456	760	691	642	442	254	109	31	30	111	286	467	632	(9)
(10)		CDD10.0	923	0	0	1	4	51	148	262	262	154	39	2	0	(10)
(11)		CDD18.3	83	0	0	0	0	1	5	34	33	10	0	0	0	(11)
(12)		CDH23.3	225	0	0	0	0	7	21	108	76	14	0	0	0	(12)
(13)		CDH26.7	15	0	0	0	0	1	2	9	3	0	0	0	0	(13)
(14)	Wind	WSAvg	3.4	4.1	3.9	3.7	3.7	3.2	2.8	2.7	2.7	2.8	3.6	4.0	4.1	(14)
(15)	Precipitation	PrecAvg	991	73	54	57	82	90	84	84	94	100	109	88	77	(15)
(16)		PrecMax	1171	120	112	139	147	156	141	137	168	192	177	138	155	(16)
(17)		PrecMin	811	38	20	8	31	33	29	38	47	38	48	41	33	(17)
(18)		PrecStd	97	21	24	28	29	31	28	28	28	35	35	25	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.7	5.5	14.2	18.2	23.6	25.5	27.8	27.1	24.9	20.6	14.2	10.2	(19)
(20)			MCWB	6.5	4.7	12.4	12.1	17.0	17.6	20.3	19.7	20.2	17.0	11.7	9.3	(20)
(21)		2%	DB	4.5	3.7	9.5	14.6	21.1	23.3	25.8	25.2	23.3	18.8	12.2	6.9	(21)
(22)			MCWB	3.8	2.9	6.7	9.3	15.1	17.2	19.7	20.0	19.8	16.2	10.2	6.0	(22)
(23)		5%	DB	3.0	2.2	6.6	12.2	19.1	21.7	24.4	24.0	22.0	17.3	10.7	5.2	(23)
(24)			MCWB	2.2	1.3	4.2	7.9	14.0	16.2	19.1	19.5	18.9	15.2	9.0	4.3	(24)
(25)		10%	DB	1.8	0.9	4.6	10.0	16.9	20.2	23.1	22.9	20.7	15.7	9.4	4.0	(25)
(26)			MCWB	0.8	-0.1	2.6	6.4	12.5	15.4	18.4	19.0	17.8	13.5	7.9	2.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	5.0	12.2	13.4	18.4	19.8	22.2	22.0	21.4	18.6	12.2	9.6	(27)
(28)			MCDB	6.7	5.2	14.2	16.6	22.6	23.1	25.5	24.9	23.4	19.8	13.8	10.2	(28)
(29)		2%	WB	4.0	3.1	7.3	10.7	16.7	18.5	21.1	21.2	20.5	16.7	10.6	6.2	(29)
(30)			MCDB	4.2	3.7	8.5	13.4	19.7	21.5	24.0	23.8	22.6	18.1	12.0	6.8	(30)
(31)		5%	WB	2.3	1.3	4.6	8.6	14.9	17.4	20.1	20.6	19.6	15.3	9.2	4.5	(31)
(32)			MCDB	2.8	2.0	6.3	11.2	17.6	20.0	23.1	22.9	21.4	16.9	10.5	5.1	(32)
(33)		10%	WB	0.9	0.1	2.7	6.6	13.3	16.4	19.2	19.9	18.6	13.7	8.0	3.0	(33)
(34)			MCDB	1.7	0.7	4.4	9.5	16.0	19.1	21.9	21.9	20.3	15.4	9.4	3.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.6	6.8	6.8	7.4	8.9	8.7	7.8	7.3	7.3	6.2	5.5	5.1	(35)
(36)			MCDBR	6.4	6.5	8.0	11.1	11.2	10.6	9.4	8.4	7.9	7.9	6.9	5.9	(36)
(37)			MCWBR	6.3	6.6	5.8	7.4	6.8	5.9	4.8	4.7	5.0	5.8	5.8	5.8	(37)
(38)		5% WB	MCDBR	6.4	6.4	7.4	9.5	9.8	8.4	7.9	6.8	6.8	7.2	6.5	5.9	(38)
(39)			MCWBR	6.4	6.6	6.2	7.7	7.0	5.8	4.8	4.6	5.1	6.1	5.9	6.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.290	0.298	0.302	0.343	0.365	0.394	0.400	0.395	0.369	0.353	0.317	0.282	(40)	
(41)		taud	2.472	2.396	2.440	2.369	2.423	2.358	2.381	2.386	2.437	2.442	2.496	2.533	(41)	
(42)		Ebn at Noon	832	889	937	917	904	874	863	855	850	812	789	809	(42)	
(43)		Edh at Noon	68	88	97	114	112	120	116	112	98	85	67	59	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.13	2.02	3.47	4.63	5.57	6.09	6.18	5.25	4.02	2.36	1.27	0.81	(44)	
(45)		RadStd	0.14	0.19	0.35	0.41	0.53	0.44	0.27	0.27	0.38	0.26	0.20	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page